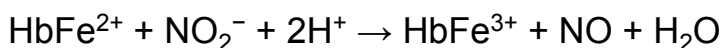


Under anaerobic conditions, hemoglobin (HbFe^{2+}) can react with nitrite to form methemoglobin (HbFe^{3+}).



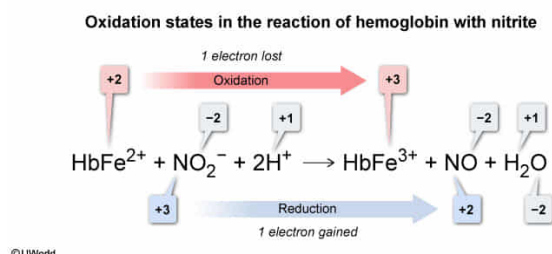
Which element is reduced in the reaction?

- ☐ A. Fe [5%]
✓ ☒ B. N [66%]
☐ C. O [11%]
☐ D. H [15%]

Correct

65% answered correctly

Explanation:



In a **redox reaction**, **oxidation and reduction** always occur together. An atom is reduced by gaining one or more electrons, but this is possible only if another atom loses the electrons (gets oxidized). As a means of tracking the gain or loss of electrons by atoms in reactions, an **oxidation number** (oxidation state) is assessed for each atom in a compound relative to the number of electrons in a neutral atom of that element.

Oxidation numbers may be assessed **using Lewis structures**; however, for compounds in which all atoms of a given element have equivalent states, values may be alternatively found by using the compound formula and **oxidation state rules**. In this alternate method, atoms with consistent oxidation states (those that rarely vary) are used to determine oxidation states for atoms that have variable states (such as metals).

Using the oxidation state rules to assign oxidation numbers to each atom in the chemical formulas of the reaction equation shows that the oxidation state of nitrogen **decreases** from +3 to +2. This shows that nitrogen is gaining a negatively charged electron and is being reduced.

(Choice A) The oxidation state of iron increases from +2 to +3. This means that iron

is *losing* a negatively charged electron and is being *oxidized* rather than being reduced.

(Choices C and D) The oxidation states of oxygen and hydrogen do not change; therefore, these elements do not undergo oxidation or reduction.

Educational objective:

Using oxidation state rules, the oxidation number of each atom in a compound formula may be determined and used to track the gain (reduction) or loss (oxidation) of electrons of an atom in a redox reaction.

Time spent:0 sec

Subject:

General Chemistry

QID:402125

System:

Interactions of Chemical Substances